

How the Proof Works

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Fix a smooth, rapidly decreasing, divergence-free initial velocity u_0 and a viscosity $\nu > 0$. Let (u, p) be the smooth Navier–Stokes solution on its maximal interval $[0, T_*)$. What matters is what the fluid would have to do if T_* were finite.

The critical height

Take a vortex as it stretches and thins. Incompressibility forces its core to narrow as it lengthens. When stretching stays ahead of diffusion, the fluid turns more rapidly around that shrinking core, so the velocity changes across a smaller distance. Its gradients can rise even while its kinetic energy remains finite.

The critical-height problem asks how a Sobolev norm reacts as that concentration moves to a finer scale. Under the Navier–Stokes rescaling

$$u_\lambda(x, t) = \lambda u(\lambda x, \lambda^2 t), \quad p_\lambda(x, t) = \lambda^2 p(\lambda x, \lambda^2 t),$$

nonlinear transport and viscous diffusion both acquire a factor λ^3 . Shrinking the scale gives neither one an automatic advantage. The Sobolev norm, however, changes by

$$\|u_\lambda(t)\|_{\dot{H}^s} = \lambda^{s-1/2} \|u(\lambda^2 t)\|_{\dot{H}^s}.$$

Below $s = \frac{1}{2}$, concentration makes the norm smaller. Above $s = \frac{1}{2}$, it makes the norm larger. At $s = \frac{1}{2}$, the scale factor disappears. The norm sees the concentration without weakening or amplifying it by a power of scale, while nonlinear transport and viscosity remain evenly matched. This is the critical height. Write

$$H(t) := \frac{1}{2} \|\Lambda^{1/2} u(t)\|_2^2, \quad \Lambda := (-\Delta)^{1/2}.$$

The shrinking core also shortens the available time. Viscous change across a width r occurs on the scale

$$\tau_\nu(r) = \frac{r^2}{\nu}.$$

Halving the width quarters that clock. Repeated again and again, this is the Zeno picture: the fluid would have to pass through smaller widths and steeper gradients in shorter intervals, all accumulating before a finite time, while its kinetic energy could remain finite.

In order for a singularity to form, velocity has to blow up. In order for that to happen, its acceleration would also have to blow up, which means its jerk has to blow up, and so on, and so on:

$$u, \quad \partial_t u, \quad \partial_t^2 u, \quad \dots$$

As the temporal order rises toward a singularity, we would expect to see less spatial regularity, not more. Navier–Stokes lets us test that expectation directly.

Velocity, acceleration, jerk

The first derivative is

$$\partial_t u = \nu \Delta u - (u \cdot \nabla)u - \nabla p.$$

Differentiate the entire equation once:

$$\partial_t^2 u = \nu \Delta \partial_t u - (\partial_t u \cdot \nabla)u - (u \cdot \nabla) \partial_t u - \nabla \partial_t p.$$

One more derivative gives

$$\begin{aligned} \partial_t^3 u &= \nu \Delta \partial_t^2 u - (\partial_t^2 u \cdot \nabla)u - 2(\partial_t u \cdot \nabla) \partial_t u \\ &\quad - (u \cdot \nabla) \partial_t^2 u - \nabla \partial_t^2 p. \end{aligned}$$

By the third line, the pattern is visible. Each time derivative acts on the entire equation, and the binomial coefficients count every place it can land in the two velocity factors. The complete pattern is

$$\begin{aligned} \partial_t^{r+1} u + \sum_{a=0}^r \binom{r}{a} ((\partial_t^a u) \cdot \nabla) \partial_t^{r-a} u + \nabla \partial_t^r p \\ = \nu \Delta \partial_t^r u. \end{aligned}$$

Set

$$V_r := \partial_t^r u, \quad Q_r := \partial_t^r p.$$

More spatial regularity, not less

Each new response exposes a spatial Sobolev height. At height s , set

$$E_s(t) := \frac{1}{2} \|\Lambda^s u(t)\|_2^2.$$

Define the nonlinear transfer by

$$\mathcal{P}_s := -\operatorname{Re} \left\langle \Lambda^s u, \Lambda^s \left((u \cdot \nabla) u \right) \right\rangle_{L^2}.$$

The viscous term moves up one Sobolev height in this energy:

$$\nu \langle \Lambda^s u, \Lambda^s \Delta u \rangle_{L^2} = -\nu \|\Lambda^{s+1} u\|_2^2 = -2\nu E_{s+1}.$$

The pressure term drops out of the global pairing because $\nabla \cdot u = 0$. Pairing the full equation gives

$$E'_s = \mathcal{P}_s - 2\nu E_{s+1}.$$

At critical height, $H = E_{1/2}$, so

$$H' = \mathcal{P}_{1/2} - 2\nu E_{3/2}.$$

Differentiate this relation and reuse the energy row at each new height. After n steps,

$$\frac{d^n H}{dt^n} = (-2\nu)^n E_{n+1/2} + \sum_{j=0}^{n-1} (-2\nu)^j \partial_t^{n-1-j} \mathcal{P}_{j+1/2}.$$

The highest term is the Sobolev energy at height $n + \frac{1}{2}$. This is the unexpected reversal: as temporal order rises, Sobolev height rises with it. We followed velocity, acceleration, jerk, and every later response expecting spatial regularity to fall away. The equation points upward through the Sobolev scale.

Apply the calculation to any temporal response. If

$$E_{r,s}(t) := \frac{1}{2} \|\Lambda^s V_r(t)\|_2^2,$$

and $\mathcal{P}_{r,s}$ collects the differentiated transport terms, then

$$E'_{r,s} = \mathcal{P}_{r,s} - 2\nu E_{r,s+1}.$$

The pressure pairing vanishes here as well because $\nabla \cdot V_r = 0$. The index r selects the temporal response; s records its spatial Sobolev height. Viscosity is the term that links one height to the next in every row.

That rise matters because, in three dimensions,

$$s > k + \frac{3}{2} \implies H^s(\mathbb{R}^3) \hookrightarrow C^k(\mathbb{R}^3).$$

For any finite k , choose a height above $k + \frac{3}{2}$. If all finite heights reach T_* , then all finite spatial derivatives reach T_* , and

$$\bigcap_{m < \infty} H^m(\mathbb{R}^3) = H^\infty(\mathbb{R}^3) \hookrightarrow C^\infty(\mathbb{R}^3).$$

The equations have exposed these heights. The finite rectangles are what keep each chosen collection controlled through T_* .

Finite rectangles

Each pair (r, m) records one concrete statement:

$$(r, m) \quad \longleftrightarrow \quad V_r = \partial_t^r u \in H_x^m.$$

Temporal order runs in one direction and spatial height in the other. Finite values N and M select a finite block. Each row is carried by an adjacent trace pair:

$$V_n \in L^2(0, T_*; H^{M+1}), \quad V_{n+1} = \partial_t V_n \in L^2(0, T_*; H^{M-1})$$

implies

$$V_n \in C([0, T_*]; H^M).$$

One row sits above the desired height, and its time derivative sits below it. Together they carry the middle trace continuously to the endpoint.

If T_* were finite, the whole-terminal rectangle theorem would supply these adjacent pairs for every finite N and M , with $0 \leq n \leq N$, across the velocity and pressure rows. That finite collection is $\mathcal{R}_{N,M}[u_0]$.

Two rectangles agree wherever they overlap because both come from the solution generated by u_0 . They assemble into $\mathcal{I}[u_0]$, whose velocity component belongs to

$$\bigcap_{r, m < \infty} H^r(0, T_*; H^m(\mathbb{R}^3)).$$

Here the temporal responses and spatial Sobolev heights meet. Every finite choice belongs to one object generated by u_0 .

The smooth endpoint

Take the zeroth temporal row of the intersection:

$$\mathcal{S}_0 := \pi_0 \mathcal{I}[u_0].$$

This row reaches the endpoint at every finite spatial height:

$$u \in C([0, T_*]; H^m(\mathbb{R}^3)) \quad \text{for every finite } m.$$

The velocity reaches one terminal value

$$u_* := u(T_*) \in \bigcap_{m < \infty} H^m(\mathbb{R}^3) = H^\infty(\mathbb{R}^3).$$

Since $H^\infty(\mathbb{R}^3) \hookrightarrow C^\infty(\mathbb{R}^3)$, u_* is smooth. The singularity picture asked the temporal responses to rise while spatial regularity disappeared. The completed intersection arrives at T_* with every finite spatial height intact.

Local smooth existence from u_* continues the fluid through $[T_*, T_* + \delta]$. Uniqueness joins that continuation to the solution on $[0, T_*]$. The proposed terminal time can be crossed, so a finite T_* cannot be maximal:

$$T_* = \infty.$$

Choose any estimate

The intersection also contains the familiar finite estimates. Restrict $\mathcal{I}[u_0]$ to any finite interval $[0, T]$, choose a target space $W^{k,p}$, and choose any finite m for which

$$H^m(\mathbb{R}^3) \hookrightarrow W^{k,p}(\mathbb{R}^3).$$

Then

$$\mathcal{I}[u_0] \longrightarrow C_t H_x^m \longrightarrow L_t^\infty W_x^{k,p}.$$

Three common choices are

$$\mathcal{I}[u_0] \longrightarrow C_t H_x^1 \longrightarrow L_t^\infty L_x^6,$$

$$\mathcal{I}[u_0] \longrightarrow C_t H_x^2 \longrightarrow L_t^\infty L_x^\infty,$$

$$\mathcal{I}[u_0] \longrightarrow C_t H_x^3 \longrightarrow L_t^\infty W_x^{1,\infty}.$$

The $W^{1,\infty}$ choice gives a familiar continuation quantity. Since

$$H^3(\mathbb{R}^3) \hookrightarrow W^{1,\infty}(\mathbb{R}^3),$$

we have

$$\|\nabla u(t)\|_{L^\infty} \leq C\|u(t)\|_{H^3},$$

and hence

$$\int_0^T \|\nabla u(t)\|_{L^\infty} dt \leq CT \sup_{0 \leq t \leq T} \|u(t)\|_{H^3} < \infty.$$

Any finite classical demand works this way: identify its derivative order and choose a Sobolev height that embeds into it. The familiar continuation estimates are read from the intersection afterward: they are derived from u_0 through the intersection, not the other way around.

The one-line recap is

$$\begin{aligned} u_0 &\longrightarrow \{(V_n, Q_n)\}_{n \geq 0} \longrightarrow \{\mathcal{R}_{N,M}[u_0]\}_{N,M < \infty} \longrightarrow \mathcal{I}[u_0] \\ &\longrightarrow \mathcal{S}_0 \longrightarrow u_* \in H^\infty \longrightarrow T_* = \infty. \end{aligned}$$